

EXAMPLE 6.3

An experiment was conducted to evaluate the effectiveness of a treatment for tapeworm in the stomachs of sheep. A random sample of 24 worm-infected lambs of approximately the same age and health was randomly divided into two groups. Twelve of the lambs were injected with the drug and the remaining twelve were left untreated. After a 6-month period, the lambs were slaughtered and the worm counts were recorded in Table 6.3:

TABLE 6.3
Sample data for treated and untreated sheep

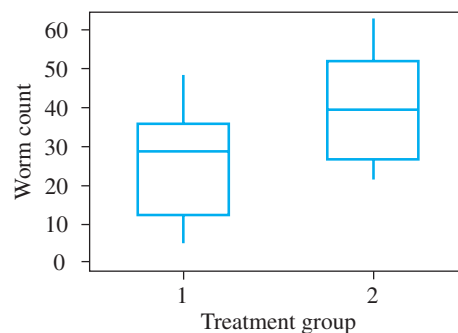
Drug-Treated Sheep	18	43	28	50	16	32	13	35	38	33	6	7
Untreated Sheep	40	54	26	63	21	37	39	23	48	58	28	39

- Test whether the mean number of tapeworms in the stomachs of the treated lambs is less than the mean for untreated lambs. Use an $\alpha = .05$ test.
- What is the level of significance for this test?
- Place a 95% confidence interval on $\mu_1 - \mu_2$ to assess the size of the difference in the two means.

Solution

- Boxplots of the worm counts for the treated and untreated lambs are displayed in Figure 6.3. From the plots, we can observe that the data for the untreated lambs are symmetric with no outliers and the data for the treated lambs are slightly skewed to the left with no outliers. Also, the widths of the two boxes are approximately equal. Thus, the condition that the population distributions are normal with equal variances appears to be satisfied. The condition of independence of the worm counts both between and within the two groups is evaluated by considering how the lambs were selected, assigned to the two groups, and cared for during the 6-month experiment. Because the 24 lambs were randomly selected from a representative herd of infected lambs, were randomly assigned to the treated and untreated groups, and were properly separated and cared for during the 6-month period of the experiment, the 24 worm counts are presumed to be independent random samples from the two populations. Finally, we can observe from the boxplots that the untreated lambs appear to have higher worm counts than the treated lambs because the median line is higher for the untreated group. The following test confirms our observation. The data for the treated and untreated sheep are summarized next.

FIGURE 6.3
Boxplots of worm counts for treated (1) and untreated (2) sheep



Drug-Treated Lambs	Untreated Lambs
$n_1 = 12$	$n_2 = 12$
$\bar{y}_1 = 26.58$	$\bar{y}_2 = 39.67$
$s_1 = 14.36$	$s_2 = 13.86$

The sample standard deviations are of a similar size, so from this and from our observation from the boxplot, the pooled estimate of the common population standard deviation σ is now computed:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{11(14.36)^2 + 11(13.86)^2}{22}} = 14.11$$

The test procedure for evaluating the research hypothesis that the treated lambs have mean tapeworm count (μ_1) less than the mean level (μ_2) for untreated lambs is as follows:

H_0 : $\mu_1 - \mu_2 \geq 0$ (that is, the drug does not reduce mean worm count)

H_a : $\mu_1 - \mu_2 < 0$ (that is, the drug reduces mean worm count)

$$\text{T.S.: } t = \frac{(\bar{y}_1 - \bar{y}_2) - D_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(26.58 - 39.67) - 0}{14.11 \sqrt{\frac{1}{12} + \frac{1}{12}}} = -2.272$$

R.R.: For $\alpha = .05$, the critical t -value for a one-tailed test with $df = n_1 + n_2 - 2 = 22$ is obtained from Table 2 in the Appendix, using $\alpha = .05$. We will reject H_0 if $t \leq -1.717$.

Conclusion: Because the observed value of $t = -2.272$ is less than -1.717 and hence is in the rejection region, we have sufficient evidence to conclude that the drug treatment does reduce the mean worm count.

- b. Using Table 2 in the Appendix with $t = -2.272$ and $df = 22$, we can bound the level of significance in the range $.01 < p\text{-value} < .025$. From the following computed output, we can observe that the exact level of significance is $p\text{-value} = .017$.

Two-Sample T-Test and Confidence Interval

Two-sample T for Treated vs Untreated

	N	Mean	StDev	SE Mean
Treated	12	26.6	14.4	4.1
Untreated	12	39.7	13.9	4.0

95% CI for μ Treated - μ Untreated: (-25.0, -1.1)

T-Test μ Treated = μ Untreated (vs <): T = -2.27 P = 0.017 DF = 22

Both use Pooled StDev = 14.1

- c. A 95% confidence interval on $\mu_1 - \mu_2$ provides the experimenter with an estimate of the size of the reduction in mean tapeworm count obtained by using the drug. This interval can be computed as follows:

$$(\bar{y}_1 - \bar{y}_2) \pm t_{.025} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$(26.58 - 39.67) \pm (2.074)(14.11) \sqrt{\frac{1}{12} + \frac{1}{12}}, \text{ or } -13.09 \pm 11.95$$

Thus, we are 95% certain that the reduction in tapeworm count through the use of the drug is between 1.1 and 25.0 worms.